QS-Net Project – Week 1 (Day-wise Task Allocation)

Team A – Dataset Analysis & Quantum-Aware Dataset Curation

Members: Interns 1, 2, 3 (Amon Koike, Mohammed Owais, Iwo Wojtakajtis)

Objective

Design a **Quantum-Aware Dataset Curation Pipeline (QADCP)** that transforms raw intrusion detection datasets into standardized, reproducible datasets optimized for Quantum Machine Learning.

Day	Tasks	Deliverables
Day 1	Study CIC-IoT2023, TON-IoT, and BoT-IoT datasets. Analyze attack categories, feature descriptions, dataset sizes, and identify common features across datasets.	Dataset comparison report, feature inventory, attack summary.
Day 2	Perform Exploratory Data Analysis (EDA): analyze missing values, duplicate records, class imbalance, outliers, feature distributions, and feature correlations.	EDA report, visualization dashboard, data quality report.
Day 3	Develop preprocessing scripts: missing-value handling, duplicate removal, categorical encoding, normalization, feature scaling, and validation.	Automated preprocessing pipeline and cleaned datasets.
Day 4	Design the **Quantum-Aware Dataset Curation Pipeline (QADCP)** covering data validation, cleaning, feature engineering, feature grouping, normalization, balancing, quantum-ready preparation, and dataset splitting.	QADCP workflow diagram and implementation.
Day 5	Develop a novel dataset curation strategy by creating **difficulty-aware zero-day splits** (Easy, Medium, Hard) based on attack similarity and clustering.	Difficulty-aware zero-day benchmark and similarity analysis.
Day 6	Create a unified feature schema across datasets. Generate Train, Validation, Calibration, Test, and Zero-Day datasets. Validate data consistency and reproducibility.	Unified dataset package and metadata documentation.
Day 7	Final validation of the complete pipeline. Package datasets, preprocessing scripts, documentation, and hand over standardized datasets to Team B and Team C.	Final QADCP, standardized datasets, documentation, and handover report.

Team B – Quantum Machine Learning

Members: Interns 4, 5 (Arjun E Naik, La Wun Nannda)

Objective

Develop the baseline Quantum Machine Learning pipeline that will later evolve into QS-Net.

Day	Tasks	Deliverables
Day 1	Install PennyLane, PyTorch, and supporting libraries. Learn quantum computing fundamentals, quantum gates, and Variational Quantum Circuits (VQCs).	Configured development environment and setup guide.
Day 2	Implement simple quantum circuits using toy datasets (Iris/XOR). Study Angle Encoding and Data Re-uploading techniques.	Working toy quantum classifier and encoding comparison notes.
Day 3	Build a baseline Variational Quantum Circuit (VQC) classifier and evaluate it on a toy dataset.	Baseline VQC implementation and performance report.
Day 4	Study the QS-Net architecture and begin implementing the quantum encoder module. Prepare the codebase for integration with Team A datasets.	Quantum encoder implementation and project structure.
Day 5	Integrate the standardized datasets received from Team A. Train the baseline VQC using the curated datasets.	Baseline quantum model trained on intrusion detection data.
Day 6	Implement quantum state generation, density matrices, quantum fidelity computation, and class prototype generation.	Fidelity computation module and prototype generator.
Day 7	Evaluate baseline performance, document observations, identify improvements required for MAQT implementation, and prepare for Week 2 development.	Baseline evaluation report and Week 2 implementation plan.

Team C – Intelligent Feature Selection

Members: Interns 6, 7 (Kevin Wong, Maral Mahmoudi Kamelabad)

Objective

Develop and compare intelligent feature-selection algorithms to identify the optimal feature subset for Quantum Machine Learning.

Day	Tasks	Deliverables
Day 1	Review literature on feature-selection techniques including Mutual Information, Random Forest Importance, BPSO, QPSO, and recent optimization-based approaches.	Literature review and algorithm comparison table.
Day 2	Implement Mutual Information and Random Forest Feature Importance. Rank features using both methods.	Baseline feature-ranking scripts and ranked feature lists.

Day	Tasks	Deliverables
Day 3	Implement Binary Particle Swarm Optimization (BPSO) for feature selection. Evaluate selected features on a classical baseline classifier.	Working BPSO implementation and preliminary results.
Day 4	Implement Quantum-Inspired Particle Swarm Optimization (QPSO). Compare it with BPSO using common evaluation metrics.	QPSO implementation and comparison report.
Day 5	Benchmark Mutual Information, Random Forest, BPSO, and QPSO using accuracy, number of selected features, runtime, and stability.	Comparative benchmark report.
Day 6	Design the **Adaptive Multi-Objective Quantum-Inspired PSO (MO-AQPSO)** framework to optimize accuracy, feature count, redundancy, and computational cost.	MO-AQPSO design document and algorithm flowchart.
Day 7	Select the best-performing feature subsets and prepare them for integration into Team B's quantum models. Document observations and future improvements.	Final feature-selection report and optimized feature subsets.

Week 1 Integration Meeting (End of Day 7)

Team A Presents

- Dataset analysis
- Quantum-Aware Dataset Curation Pipeline (QADCP)
- Standardized Train, Validation, Calibration, Test, and Zero-Day datasets
- Dataset documentation

Team B Presents

- Baseline quantum machine learning pipeline
- Quantum encoding methods
- Baseline VQC performance
- Prototype generation progress

Team C Presents

- Comparative study of feature-selection algorithms
- BPSO and QPSO implementations
- Proposed MO-AQPSO framework
- Recommended feature subsets

Week 1 Success Criteria

By the end of Week 1, the project should have:

- ✅ A fully reproducible Quantum-Aware Dataset Curation Pipeline (QADCP).
- ✅ Standardized intrusion detection datasets ready for experimentation.
- ✅ A working baseline Variational Quantum Circuit (VQC).
- ✅ Implementations of Mutual Information, Random Forest, BPSO, and QPSO feature-selection methods.
- ✅ Optimized feature subsets ready for quantum model integration.
- ✅ Complete documentation to enable a smooth transition into Week 2, where the focus will shift to implementing the core QS-Net methodology, including Margin-Aware Quantum Training (MAQT), prototype generation, conformal prediction, and robustness evaluation.